

VPA A-Level CS — AS Level Semester 1 Course Overview

VPA A-Level Computer Science — AS Level Semester 1

Course Overview: Theory Fundamentals & Programming Introduction

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This Overview

Weeks 1 – 9: Information Representation & Communication

Weeks 10 – 18: Hardware, System Software & Security

Assessment Pack

Course Code: CIE A-Level Computer Science (9618)

Semester Duration: 18 weeks (September – January)

Lesson Duration: 40 minutes

Target Students: Year 11 equivalent (Post-IGCSE), EAL (Mandarin L1), mixed ability

Papers Covered: Paper 1 (Theory Fundamentals, 1h30, 75 marks) + Paper 2 intro

1. Course Summary and Philosophy

This semester establishes the theoretical foundations of computer science essential for success at A-Level. Building upon the strong IGCSE Computer Science (0478) foundation already established at VPA, students will deepen their understanding of how computers represent information, communicate across networks, and process data at the hardware level. The semester introduces programming fundamentals that will be developed intensively in Semester 2.

教学理念 (Jiàoxué lǐniàn): We employ a bilingual scaffolding approach where core concepts are introduced in English with Mandarin annotations, gradually reducing L1 support as students build academic confidence. Every lesson incorporates EAL strategies including vocabulary pre-teaching, visual anchors, and structured speaking frames.

2. Student Profile and Implications

Characteristic	Implication for Teaching
Mandarin L1, English academic language developing	Bilingual glossary for every topic; sentence starters for discussions; visual concept maps
Mixed ability (IGCSE grades A*–C range)	Bronze/Silver/Gold differentiation in every lesson; pre-assessment diagnostics
Strong IGCSE foundation in binary/hex, basic programming	Bridge document identifies prior knowledge vs. new depth; avoid unnecessary repetition
University aspirations (UK/US/China)	Link concepts to university CS courses; emphasise exam technique
40-minute lesson blocks	Pacey lessons with clear starter/main/plenary structure; homework extends consolidation

3. Eighteen-Week Curriculum Map

Week	Topic (Syllabus Ref)	Paper	Key Content	Assessment
1	1.1 Information Representation — Number Systems	P1	Binary, denary, hex conversion; two's complement	Diagnostic test
2	1.1 — Text, Images, Sound	P1	ASCII, Unicode, bitmap/vector, sampling, resolution	Exit ticket
3	1.1 — Compression	P1	Lossy vs lossless; RLE, Huffman, dictionary	Quiz
4	1.2 Communication — Networks	P1	Client-server, P2P, LAN/WAN, protocols	Worksheet
5	1.2 — Internet Technologies	P1	WWW, email, FTP, packet switching, TCP/IP overview	Practical task
6	1.2 — IP Addressing & Protocols Deep Dive	P1	IPv4/IPv6, DNS, MAC addresses, protocol layers	Mid-unit test
7	1.3 Hardware — CPU Architecture	P1	Von Neumann, registers, FDE cycle, buses	Practical simulation
8	1.3 — Memory & Storage	P1	RAM/ROM/cache, memory hierarchy, secondary storage	Comparison table
9	Consolidation Week & Half-Term Assessment	P1	Review weeks 1–8; Paper 1 past paper section A	Half-Term Exam
10	1.4 Processor Fundamentals — Machine Architecture	P1	ALU, CU, registers, instruction sets	Lab worksheet
11	1.4 — Assembly Language	P1	Mnemonics, addressing modes, simple programs	Programming task
12	1.5 System Software — Operating Systems	P1	OS functions, memory management, scheduling	Research poster
13	1.5 — Translators	P1	Compiler, interpreter, assembler; lexical/syntax analysis	Comparison essay
14	1.6 Security — Threats & Malware	P1	Virus, worm, trojan, phishing, social engineering	Case study
15	1.6 — Encryption & Security Protocols	P1	Caesar, Vernam, SSL/TLS, firewalls, passwords	Practical encryption
16	1.7 Ethics & Ownership	P1	Licensing, AI ethics, environmental issues, DPA	Debate preparation

Week	Topic (Syllabus Ref)	Paper	Key Content	Assessment
17	Programming Introduction — Pseudocode & Flowcharts	P2	Algorithm design, structure diagrams, IPO charts	Design task
18	End-of-Semester Assessment & Review	P1/P2	Full Paper 1 mock; programming diagnostic	Mock Exam

4. Assessment Calendar

Date	Assessment	Format	Marks	Weighting
Week 1	Baseline Diagnostic	30-min MCQ + short answer	30	Formative
Week 3	Information Representation Quiz	20-min problem set	20	Formative
Week 6	Mid-Unit Test (Communication)	45-min structured questions	40	10%
Week 9	Half-Term Examination	1h Paper 1 Section A	35	15%
Week 12	System Software Essay	Timed 20-min essay	15	5%
Week 15	Security Case Study Test	30-min scenario questions	25	10%
Week 18	End-of-Semester Mock	Full Paper 1 (1h30)	75	25%
Ongoing	Lesson Exit Tickets	5-min reflective questions	n/a	Formative

5. Standard Lesson Structure (40 Minutes)

0–5 min: Starter / Do Now

- ☐ Retrieval practice question from previous lesson
- ☐ Vocabulary review (bilingual flashcards)
- ☐ Quick diagnostic MCQ

5–25 min: Main Activity

- ☐ Concept introduction with visual model
- ☐ Worked example (teacher modelling)
- ☐ Guided practice (pairs/groups)
- ☐ Independent practice with Bronze/Silver/Gold tasks

25–35 min: Consolidation

- ☐ Plenary check (exit ticket or whiteboard response)
- ☐ Common misconception addressed
- ☐ Preview of next lesson

35–40 min: Set Homework

- ☐ Exam-style question (differentiated)
- ☐ Vocabulary consolidation

☐ Reading/preparation for next lesson

6. EAL Support Checklist

Every lesson must include:

- ☐ Bilingual keyword list (English / 中文 / Pinyin) displayed
- ☐ Visual anchor (diagram, image, concept map) for abstract concepts
- ☐ Sentence starters for speaking and writing tasks
- ☐ Cognates highlighted (e.g., "communication" = 通信 tōngxìn)
- ☐ Checking for understanding using non-verbal methods (thumbs, mini-whiteboards)
- ☐ Word banks for extended writing
- ☐ Pre-teaching of 5 key terms before each new topic
- ☐ Bilingual glossary section in notes

7. Differentiation Framework

Tier	Description	Lesson Application
Bronze	Core knowledge recall; scaffolded tasks; sentence frames	Complete conversion tables with hints; label diagrams; match definitions
Silver	Application and explanation; some independent work	Explain processes in own words; compare/contrast; solve multi-step problems
Gold	Analysis, evaluation, synthesis; extension tasks	Evaluate methods; design solutions; link to real-world contexts; exam 9-markers

8. Resource Requirements

Resource	Purpose	Frequency
CIE 9618 syllabus (2024 - 2026)	Content accuracy	Constant reference
Past papers (2010 - 2024)	Question style familiarity	Weekly
Online assembly simulator (e.g., Little Man Computer)	Assembly language visualisation	Weeks 10 - 11
Binary/hex conversion tools	Skill practice	Weeks 1 - 3
Wireshark (packet analysis)	Network protocol visualisation	Weeks 4 - 6
Bilingual dictionaries (CS-specific)	Vocabulary support	Every lesson
VPA CS Student Handbook (IGCSE revision)	Bridge from prior knowledge	Week 1 diagnostic

9. Key Bilingual Glossary (50 Terms)

Binary 二进制 (èr jìnzhì) — Base-2 number system using 0 and 1

Hexadecimal 十六进制 (shíliù jìnzhì) — Base-16 number system (0 - 9, A - F)

Two's complement 补码 (bǔmǎ) — Method for representing negative binary numbers

ASCII 美国信息交换标准代码 (Měiguó xìnxī jiāohuàn biāozhǔn dàimǎ) — 7-bit character encoding

Unicode 统一码 (tǒngyī mǎ) — Universal character encoding standard

Pixel 像素 (xiàngsù) — Smallest unit of a digital image

Resolution 分辨率 (fēnbiànlǜ) — Number of pixels per dimension

Sample rate 采样率 (cǎiyàng lǜ) — Number of audio samples per second

Bit depth 位深度 (wèi shēndù) — Bits per sample in audio

Lossy compression 有损压缩 (yǒusǔn yāsuō) — Compression with data loss

Lossless compression 无损压缩 (wúsuǎn yāsuō) — Compression without data loss

Run-length encoding 行程长度编码 (xíngchéng chángdù biānmǎ) — RLE compression

Client-server 客户机-服务器 (kèhùjī-fúwùqì) — Network architecture model

Peer-to-peer 点对点 (diǎn duì diǎn) — Decentralised network model

LAN 局域网 (júyù wǎng) — Local Area Network

WAN 广域网 (guǎngyù wǎng) — Wide Area Network

Packet switching 分组交换 (fēnzǔ jiāohuàn) — Data transmission method

Protocol 协议 (xiéyì) — Rules for data communication

TCP/IP 传输控制协议/网际协议 (chuánshū kòngzhì xiéyì/wǎngjì xiéyì) — Internet protocol suite

DNS 域名系统 (yùnmíng xìtǒng) — Domain Name System

IP address IP地址 (IP dìzhǐ) — Unique network identifier

MAC address MAC地址 (MAC dìzhǐ) — Hardware address

Router 路由器 (lùróuqì) — Device forwarding packets between networks

Switch 交换机 (jiāohuànqì) — Device connecting devices in LAN

Firewall 防火墙 (fánghuǒqiáng) — Network security barrier

CPU 中央处理器 (zhōngyāng chǔlǐ qì) — Central Processing Unit

ALU 算术逻辑单元 (suànshù luójí dānyuán) — Arithmetic Logic Unit

Control unit 控制单元 (kòngzhì dānyuán) — Directs processor operations

Register 寄存器 (jìcúnqì) — Fast storage location in CPU

Program counter 程序计数器 (chéngxù jìshùqì) — Holds address of next instruction

Fetch-decode-execute 取指-译码-执行 (qǔzhǐ-yì mǎ-zhí xíng) — CPU instruction cycle

Bus 总线 (zǒngxiàn) — Data pathway

RAM 随机存取存储器 (suíjī cúnqǔ cúncǔqì) — Random Access Memory

ROM 只读存储器 (zhǐdú cúncǔqì) — Read Only Memory

Cache memory 高速缓存 (gāosù huǎncún) — Fast CPU memory

Machine code 机器码 (jīqì mǎ) — Binary instructions

Assembly language 汇编语言 (huìbiān yǔyán) — Low-level symbolic code

Compiler 编译器 (biānyì qì) — Converts high-level to machine code

Interpreter 解释器 (jiěshì qì) — Executes code line-by-line

Operating system 操作系统 (cāozuò xìtǒng) — System software managing resources

Memory management 内存管理 (nèicún guǎnlǐ) — OS allocation of RAM

Multitasking 多任务处理 (duō rènwù chǔlǐ) — Running multiple processes

- Virus 病毒 (bìngdú) — Malicious self-replicating software
- Encryption 加密 (jiāmì) — Encoding data for security
- Plain text 明文 (míngwén) — Unencrypted data
- Cipher text 密文 (mìwén) — Encrypted data
- Digital signature 数字签名 (shùzì qiānmíng) — Authentication mechanism
- Open source 开源 (kāiyuán) — Publicly available source code
- Artificial intelligence 人工智能 (réngōng zhìnéng) — Machine intelligence

10. Pseudocode Reference (Preview for Semester 2)

CIE Pseudocode Conventions (9618):

```
DECLARE :  
  
CONSTANT =  
  
INPUT  
  
OUTPUT  
  
IF THEN ... ELSE ... ENDIF  
  
FOR = TO ... ENDFOR  
  
WHILE DO ... ENDWHILE  
  
REPEAT ... UNTIL  
  
PROCEDURE () ... ENDPROCEDURE  
  
FUNCTION () RETURNS ... ENDFUNCTION
```

These constructs will be taught intensively in Semester 2.

11. Past Paper Reference Guide

Topic	Recommended Past Papers	Question Types
Number Systems	9618/11 MJ22 Q1, 9618/12 ON21 Q2	Conversion, two's complement
Images/Sound	9618/11 MJ23 Q3, 9618/12 ON22 Q1	Calculation, comparison
Compression	9618/11 ON22 Q4, 9618/12 MJ22 Q3	RLE, Huffman encoding
Networking	9618/11 MJ22 Q5, 9618/12 ON21 Q4	Protocol explanation, diagram
CPU Architecture	9618/11 MJ23 Q2, 9618/12 ON22 Q2	FDE cycle, register functions
Assembly	9618/11 ON22 Q3, 9618/12 MJ23 Q4	Trace table, code writing
System Software	9618/11 MJ22 Q4, 9618/12 ON21 Q5	Comparison, evaluation
Security	9618/11 ON22 Q5, 9618/12 MJ22 Q5	Scenario analysis, encryption
Ethics	9618/11 MJ23 Q6, 9618/12 ON22 Q6	9-mark evaluation essays

Teacher Note: All past paper references use the CIE 9618 specification. Where 9618 papers are limited, 9608 (legacy specification) papers may be used for additional practice with teacher vetting for syllabus alignment. Ensure all questions used are from the 2024–2026 syllabus where possible.